

ClimateScope – Milestone 3 Report

Project Title: Interactive ClimateScope Weather Analysis Dashboard

Technology Used: Streamlit, Python, Plotly

Dataset: Cleaned Global Weather Dataset

1. Introduction

Milestone 3 focuses on visualization development and interactivity. The objective of this milestone is to build an interactive dashboard that allows users to explore global climate patterns, analyze trends, identify extreme weather events, and gain actionable insights through dynamic visualizations.

2. Tools and Technologies

- Streamlit for interactive dashboard development
- Plotly for interactive charts and geographic visualizations
- Pandas and NumPy for data processing and aggregation

3. Dashboard Architecture

The ClimateScope dashboard is structured into multiple pages using Streamlit navigation. Sidebar-based filtering allows users to dynamically control country selection, date range, climate metric, and extreme event thresholds. The main content area updates instantly based on user inputs.

4. Implemented Pages and Features

Executive Dashboard: KPI cards showing average, maximum, minimum values, latest weather snapshot, and a global interactive map.

Statistical Analysis: Scatter plot, correlation heatmap, descriptive statistics table, histogram, and box plot.

Climate Trends: Monthly trend line charts and country-wise distribution plots.

Extreme Events: Detection of extreme weather events using thresholds, frequency analysis, and top extreme event tables.

Help Section: User guide explaining navigation and dashboard usage.

5. Interactivity and User Experience

The dashboard supports hover tooltips, zooming and panning, animated maps, dropdown-based filtering, date range selection, and data export functionality. A clean and responsive layout ensures usability across screen sizes.

6. Data Processing and Handling

Data preprocessing includes handling missing values, converting date columns to datetime format, creating monthly aggregations, and grouping data at country and temporal levels for analysis.

7. Conclusion

The ClimateScope dashboard successfully fulfills all Milestone 3 requirements. It delivers a near-complete interactive visualization system with advanced interactivity, meaningful insights, and a professional user interface. The dashboard is ready for evaluation and further enhancement.